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ARCH PRESENTS

2ND EDITION OF

HACK-O-WEEN

10HR HACKATHON

Crowd Whisperers

Crowd-crush early warning

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Team Name: **Crowd Whisperers**

Track Selected: **Voodoo Vision** (Computer Vision)

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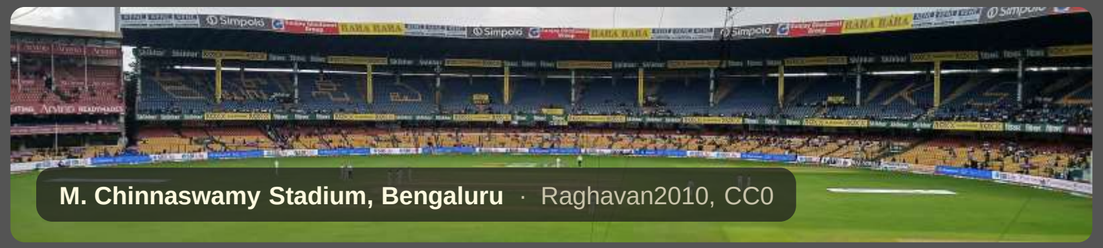
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Problem Statement

11
deaths in the Chinnaswamy Stadium crush, June 2025

2–3 lakh
fans at a stadium built for 35,000, per Karnataka's CM

186
deaths in five Indian crowd crushes in just 12 months



M. Chinnaswamy Stadium, Bengaluru · Raghavan2010, CC0



Maha Kumbh crowd, Prayagraj · Michael T. Balonek, CC BY-SA 4.0

The problem we're solving

Organisers can't see a crush building, and help reaches the wrong spot too late.

- No one measures crowd density live
- A crush builds in minutes, with no alarm
- Help is slow to reach someone who falls

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Novelty

1 How it's different

- ✗ Today: guards watch CCTV and guess crowd size
- ✓ **Predicts when a zone will turn critical, minutes ahead**

- ✗ Today: people-counters miss heads in packed crowds
- ✓ **Switches to a density-map model when the crowd packs in**

- ✗ Today: alarms stop at the control room
- ✓ **Sends the nearest help station, with location and photo**

2 How it solves the problem

Warns early

"Gate 3 critical in 6 min", from live gate inflow

Sends help fast

Nearest medical or police post gets the alert; escalates if no reply

Moves the crowd

Suggests which gate to open; PA speaks in Kannada, Hindi, English

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Brief About the Idea

Poltergeist predicts crowd crushes, spots incidents and sends the nearest help, using cameras a venue already has.

Predict

Live heatmap + gate inflow: "critical in 6 min"

Detect

Crush risk, a person down, or a panicked scatter

Dispatch

Banshee alerts the nearest help post, escalates after 60 s

Redirect

Suggests the gate to open; PA announces in 3 languages

Private by design: faces are blurred on screen and only counts are stored, never identities

Built for: stadiums · concerts · temples and festivals · railway stations · college fests

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Proposed Solution

1 Capture
Live CCTV or phone video, faces blurred

→

2 Count
YOLO if sparse, CSRNet if packed

→

3 Predict
People per m² + minutes to critical

→

4 Detect
Surges and scatter (optical flow), falls (pose)

→

5 Respond
Nearest help post, gate advice, PA

Risk levels per zone

SAFE
under 4 /m²

WARNING
4 to 6 /m²

CRITICAL
6+ /m² or a surge

Based on crowd scientist Prof. G. Keith Still's guidance; tunable per venue.

Our 10-hour build, 4 people

- **P1** Video, YOLO + pose, face blur
- **P2** Density model, heatmap, prediction
- **P3** Optical flow + risk engine
- **P4** Dashboard, dispatch, PA voice

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Technologies Used

Vision models

- YOLOv8 people + pose (falls)
- CSRNet for dense crowds

Video and motion

- OpenCV: zones, face blur
- Farneback optical flow

Prediction

- Python, PyTorch, NumPy
- Inflow trend → time to critical

Dashboard

- Streamlit live control room
- Venue map of help posts

Dispatch + PA

- Telegram bot + Accept button
- gTTS voice in 3 languages

Data and testing

- ShanghaiTech for accuracy
- UMN + our clips for surges



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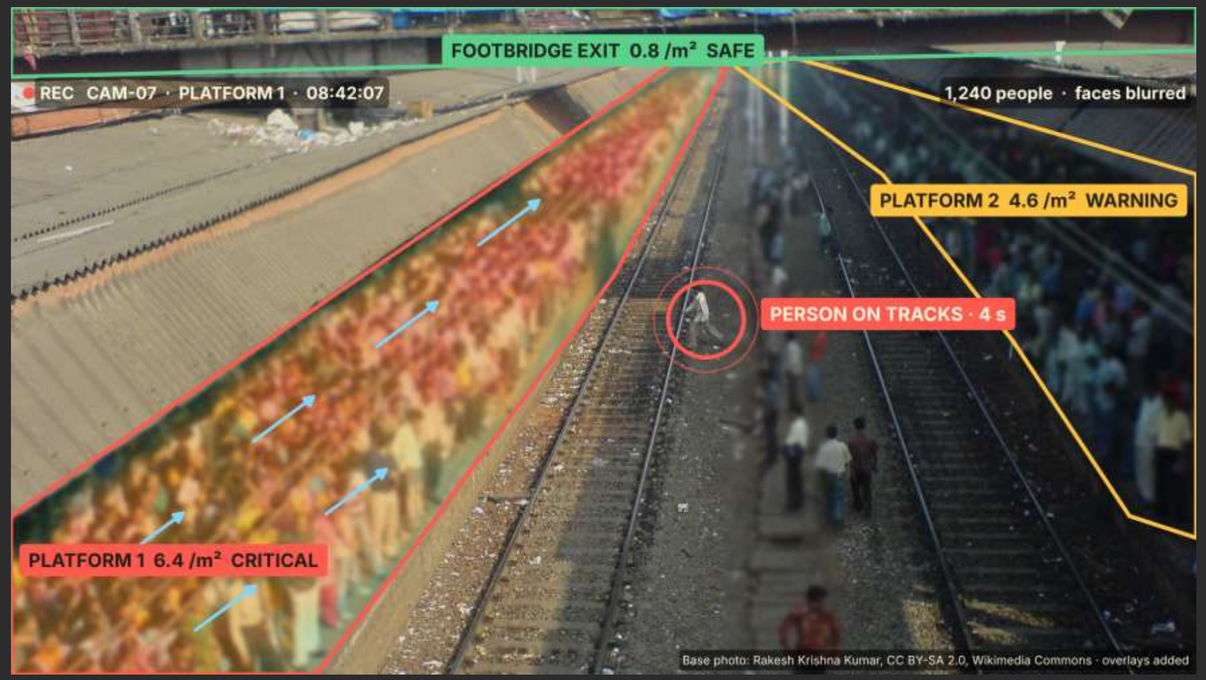
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Wireframe: Live Control Room

Poltergeist · Live crowd monitor

3 cameras live · faces blurred · 08:42



ZONES

Platform 1	6.4 /m²	CRITICAL
Platform 2	4.6 /m²	CRITICAL IN 6 MIN
Footbridge	0.8 /m²	SAFE

CRITICAL · Platform 1

6.4 /m², crowd surging, person on tracks

Accepted: RPF Post 2, 60 m away

Do now: route to footbridge; PA is on

Also send medical | Acknowledge

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Thank You!

Crowd Whisperers · Voodoo Vision







